# Multimodal AI Teaching Copilot: Live Slide Generation from Lecture Speech

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***Abstract*—Prepared slides cannot show what a teacher adds while explaining, and a large language model (LLM) that regenerates slides from a growing transcript rewrites what students are reading. This paper presents a teaching copilot that builds a structured, continuity-aware slide deck on the classroom projector from the teacher's live speech. Locally transcribed speech is sent in bounded units to an LLM that only returns a schema-validated interpretation; a deterministic planner turns it into slide operations that only extend the deck, so the LLM cannot rewrite what students have read, and a transcript-coverage guard restores facts the LLM omits. Across 36 held-out lectures (409 min), run on one laptop with 4 GB of graphics memory, the median delay from speech to slide was 11.2 s. On an annotated subset of 18 lectures, slide changes matched topic boundaries with precision 0.78, 86.6 % of key facts reached a slide, 0.9 % of statements were unsupported by the speech, and five teachers gave mean ratings of 3.87–4.17 of 5 on four criteria. On the same units, an LLM-controlled baseline rewrote or removed 38.2 % of the statements students had already seen, against 2.0 % (lower in all 18 lectures, $p < 0.001$), and a replay ablation showed that opening a slide per concept lowers precision to 0.37.**

***Index Terms*—Automatic slide generation, educational technology, large language models, live systems, speech recognition, human-in-the-loop**

## I. Introduction

Slides prepared before a lecture fix its order in advance, so the examples and answers a teacher adds while explaining never reach the screen. Narration paired with relevant pictures helps learners more than narration by itself [1], provided the visuals add no extraneous load [2], [3]; yet no teacher can draw such visuals while teaching.

Existing tools solve only parts of this problem: live captions show verbatim text without the structure of the lesson [4]; slide generators work offline from a finished document or prompt [5]–[11]; and a large language model (LLM) [12], [13] applied to a growing transcript rewrites slides while students read them and adds untaught content [14]. That leaves three gaps. What is projected does not follow what is being taught (P1). Live speech is never turned into a structured visual lesson (P2). And LLM output is neither stable nor faithful to the teacher's words (P3).

The Multimodal AI Teaching Copilot was built to close these gaps. It listens to the teacher and, while the teacher is still talking, puts typed layouts, typeset formulas, diagrams and retrieved images on the projector. The contributions are: (1) incremental slide generation from live speech; (2) an extend-only update policy: the LLM only labels bounded, schema-validated units, and a deterministic planner executes only operations that add to the deck, so the LLM cannot rewrite what students have read; (3) a transcript-coverage guard that restores facts the LLM omits, and a formula guard; and (4) a held-out evaluation with annotation, teacher ratings, a replay ablation and a paired comparison with an LLM-controlled baseline.

## II. Related Work

### A. Multimedia Learning and Lecture Speech

In Mayer's experiments, learners given text with matching pictures outperformed those given text only [1]; Sweller attributes such effects to how little a learner can hold in mind at once [2], [3], and decorative pictures reduced test scores in PowerPoint lectures [15]. The system therefore uses short typed layouts and adds an image only for a concrete object that is taught, vetted with CLIP, which Radford *et al.* trained on image–caption pairs so that any picture can be scored against a text prompt [16].

Whisper by Radford *et al.* [17], run with 8-bit weights by faster-whisper [18], and Silero VAD [19] form the speech front end. Kushalnagar *et al.* showed that verbatim real-time captions are hard for students to follow [4], and Gonzalez *et al.* found that LLM summaries of recorded lectures improved quiz scores [20]; both produce text, not a structured visual lesson (P2).

### B. Segmentation, Slide Generation and LLMs

TextTiling by Hearst places subtopic boundaries where the similarity of adjacent blocks drops [21]; Sentence-BERT by Reimers and Gurevych compares sentence embeddings by cosine similarity [22], and Wang *et al.* distilled it into MiniLM [23]. These methods segment a complete transcript offline; here the similarity drop is one online signal. Boundaries are scored with the $P_k$ measure of Beeferman *et al.* [24], and fact coverage with the word-recall overlap that ROUGE-1 by Lin uses for summaries [25].

PPSGen picks the sentences of a paper that go on slides with integer linear programming [7]. D2S writes text for each slide title from a document [6], and DOC2PPT adds figures as well [5]. LLM systems go further: Xie *et al.* ground educational slides in retrieved sources [8], SlideBot splits the planning across agents [9], PPTAgent edits reference slides [10], and AutoPresent writes slides as code [11]. All of them build the whole deck at once from clean written input; a lecture arrives sentence by sentence as noisy speech (P1, P3).

TABLE I
COMPARISON OF THE PROPOSED SYSTEM WITH PRIOR SYSTEMS

| System | Input | Produced | Topic structure | Faithfulness to source | Slide sequence |
|---|---|---|---|---|---|
| Live captions [4],[17] | Speech | Live | None (text stream) | Verbatim text | None |
| Lecture summaries [20] | Recorded lecture | Offline | None (summary) | LLM summary | None |
| PPSGen, D2S, DOC2PPT [5]–[7] | Paper / document | Offline | Document sections | Extracted from document | Whole deck at once |
| LLM slide generators [8]-[11] | Topic, document | Offline | LLM outline / plan | Retrieved sources or none | Whole deck at once |
| **Proposed system** | **Live speech** | **During lecture** | **Online topic / subtopic** | **Coverage + formula guards** | **Extend-only + teacher** |

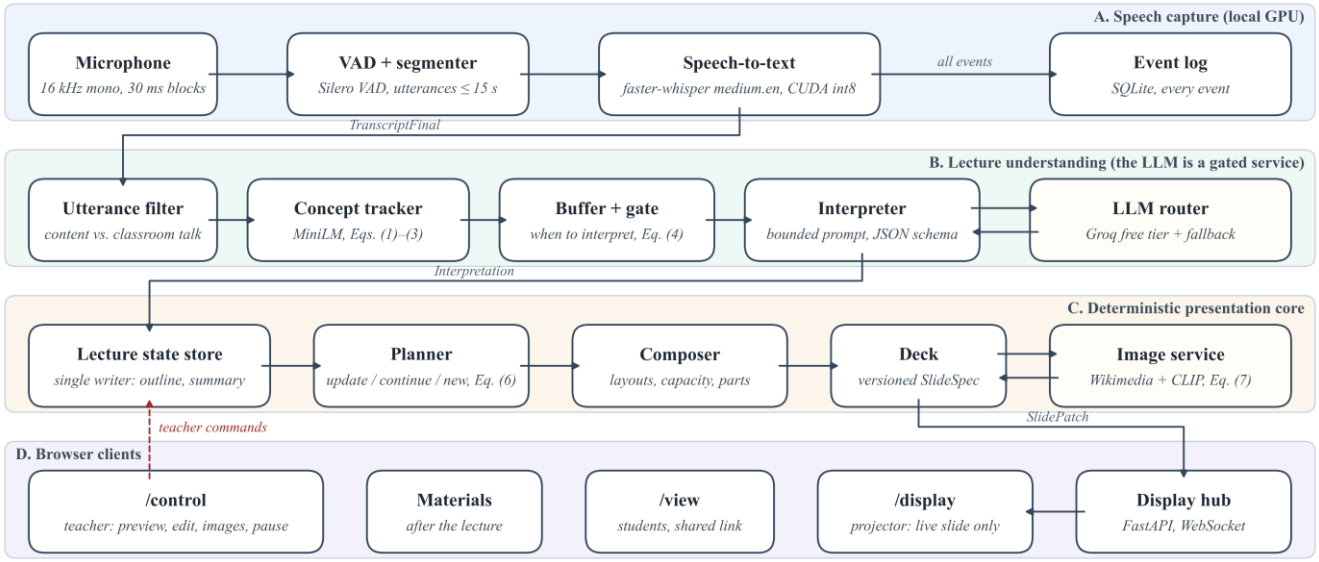



Fig. 1. Implemented architecture. Lanes A–C share one Python process; only the interpreter talks to the LLM.

Kasneci *et al*. argue that teachers must stay in charge when LLMs are used in class [12]; here the teacher can override every automatic decision. Ji *et al.* surveyed hallucination [14], and Willard and Louf constrained generation to a JSON schema [26], which fixes the format but not the facts, so the output is also checked against the transcript. Open-weight models, gpt-oss-120b [13] and the Qwen3 family [27], serve as interpreters on a free hosted tier; generated quizzes follow the review of Kurdi *et al.* [28]. The agreement between the annotators is evaluated using Cohen's kappa [29] and AC1 [30] by Gwet, the latter of which is especially suited for the situations with label dominance; and teacher grades with ICC(2,1) by Shrout & Fleiss [31].

These methods are compared in Table I. But none of them takes continuous speech as input and adjusts the slides according to the pace of the lecture. This is addressed in this paper.

## III. METHODOLOGY

### *A. System Overview*

The pipeline is illustrated in the diagram below. Initially, the speech is recorded and analyzed. The fillers are filtered from the speech and a concept tracker follows the topic. Thereafter, the gate is applied, in which the processing is delayed till the end of the sentence, which is then sent to the LLM for comprehension with some context. The modules of planning and composing are deterministic and the response is transformed into slide operations and layouts, which are then sent to the projector as well as the teachers and students. There are four constraints identified at the outset: The LLM produces only labels and performs no operations on the slides; It receives only parts of the transcript; New concept does not necessarily imply new slide; User interfaces only send commands.

All components run in one Python process using the asyncio library and exchange typed events; every event is recorded in an SQLite database, from which lectures are replayed for evaluation and materials. Recognition runs on the GPU, MiniLM and CLIP on the CPU (ONNX), and the LLM on the Groq free tier (openai/gpt-oss-120b [13]; qwen/qwen3.8-27b of the Qwen3 family [27] when that quota is spent; Oct. 2026), as a 7B-class local model does not fit beside the speech model in 4 GB. Of the 1,619 held-out units, 71.6 % were interpreted by the first model, 24.0 % by the second and 4.4 % by the deterministic fallback.

### *B. Speech Capture and Concept Tracking*

Silero VAD [19] segments 16 kHz audio (utterances up to 15 s), and faster-whisper [18] with Whisper medium.en [17] (int8, beam size 1, keyword prompt) transcribes them. A deterministic filter removes classroom management, meta talk and fillers. Each content utterance is embedded with MiniLM [23]; with the L2-normalized embedding $e_t$ and concept centroid $c_{t-1}$, the shift score is the cosine distance $s_t = 1 - e_t \cdot c_{t-1}$ used with Sentence-BERT [22], and the centroid is the

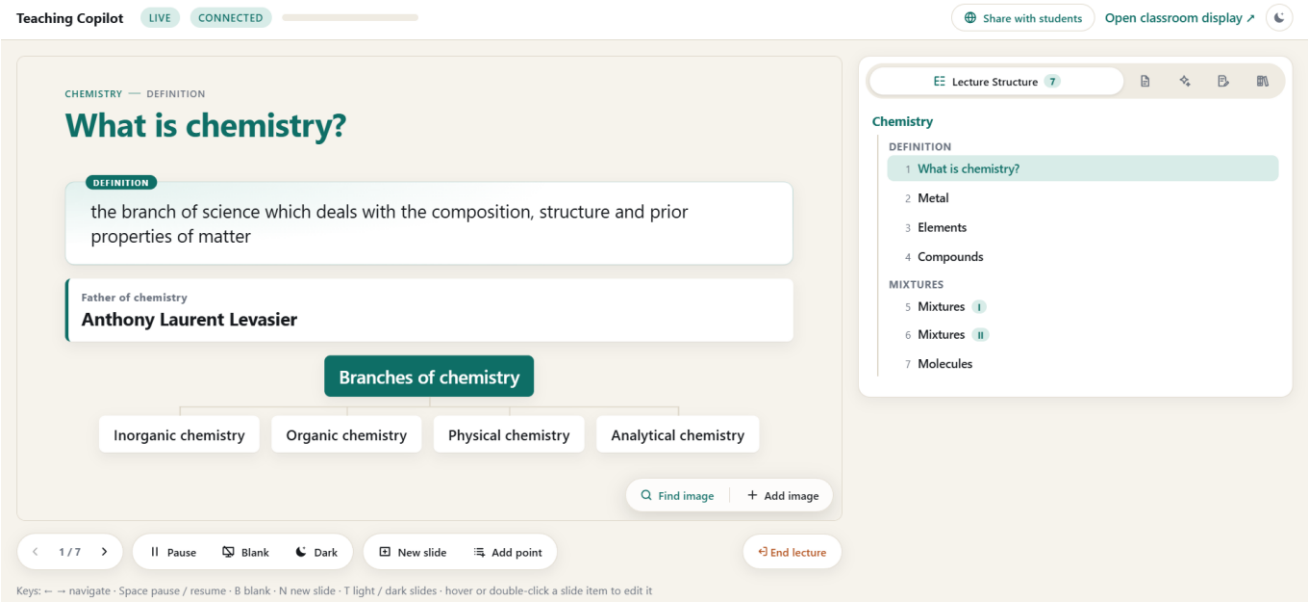



Fig. 2. Teacher control view (/control) in a live chemistry lecture: the projected slide as an editable preview with image tools and lecture controls (left) and the lecture structure tree built by the system, topic → facet → slides, with overflow parts I and II (right).

exponentially weighted mean $c_t = \mathrm{norm}((1-\alpha)c_{t-1} + \alpha e_t)$ with $\alpha = 0.35$, raised to 0.6 at a boundary so it re-seeds on the new concept. Following the similarity-drop criterion of TextTiling [21], a boundary is signalled when

$$b_t = \kappa_t \vee (s_t \geq \tau), \tau = 0.75 \quad (1)$$

where $\kappa_t$ marks a cue phrase ("moving on to", "next"); $\tau$ is high because shift scores within one facet were 0.3–0.6.

### C. Gate and Token Budget

Content accumulates in a buffer until a rule-based gate fires. With $p_t$ the silence after a finished sentence, $w_t$ the buffered words and $a_t$ the age of the oldest pending line (s),

$$G_t = (p_t \geq 1.2 \wedge w_t \geq 6) \vee w_t \geq 40 \vee b_t \vee a_t \geq 12 \quad (2)$$

with an 8 s floor between calls, set by the free quota (200,000 tokens per key and model per day).

### D. Bounded Interpretation and Transcript-Grounding Safeguards

Each prompt carries the subject and level, an outline of at most 150 tokens, the current slide with its items numbered, a rolling summary (at most 120 tokens) and the new lines, under 3,000 tokens in all (temperature 0.2). The answer is JSON: topic and subtopic, how the unit relates to what came before (same concept, elaboration, sub-concept, sibling, new topic or digression), typed content, an image hint, and *concerns*, i.e. statements the model suspects are wrong. Pydantic checks it; if one repair attempt also fails, a deterministic fallback just shows the spoken sentences as points. Because the model sometimes drops a fact, a coverage guard compares every content sentence u of six or more words with the output O. With $W(\cdot)$ the set of word stems and a recall-style overlap as in ROUGE-1 [25],

$$|W(u) \cap W(O)| \leq \tfrac{1}{3}|W(u)| \Rightarrow u \text{ is shown as said} \quad (3)$$

If the model has left out a sentence, that sentence is put back on the slide. Formulas have a separate guard, which flags any changed token. A correction is applied only when its confidence is at least 0.75 for a factual error or 0.4 for a mis-hearing. Otherwise the slide keeps what was said, and the teacher gets a card with both versions in the control view. The guards keep the slide faithful to the transcript; they do not prove a statement correct.

### E. Slide Planner

For each interpretation the planner picks one operation: update the working slide, continue on a new slide of the same topic, open a new topic slide, retitle, or do nothing. No operation lets the LLM rewrite or delete something already shown. Shown text changes in three cases only: the composer merges an elaboration into the item it elaborates, a mis-hearing is corrected, or the teacher edits it. One unit can be misread, so a topic change has to be confirmed:

$$C_n = B_n \vee \left(\bar{s}_n^{\max} \geq \tau\right) \vee \left(T_n \equiv \tilde{T}_{n-1}\right) \quad (4)$$

where $B_n$ is true if a boundary (1) occurred in unit $n$, $\bar{s}_n^{\max}$ is its largest shift score smoothed over two utterances, $T_n$ its topic and $\tilde{T}_{n-1}$ an unconfirmed candidate from the previous unit. Unconfirmed content is shown tentatively and moved once confirmed. A minimum dwell of 15 s (6 s for the next part of a full slide) prevents rapid changes, near-duplicates (similarity $\geq 0.85$) are merged, and teacher edits are never overwritten. All thresholds were set by hand on D1 and are tested by ablation in Section IV.

### F. Composer and Image Retrieval

Items become definition cards, key points, fact tiles, process flows, comparison tables, timelines, cause-effect chains, classification trees or formulas. A full slide continues in a further part (I, II, …); text is never shrunk to fit, and spoken mathematics is typeset with KaTeX. For a concrete

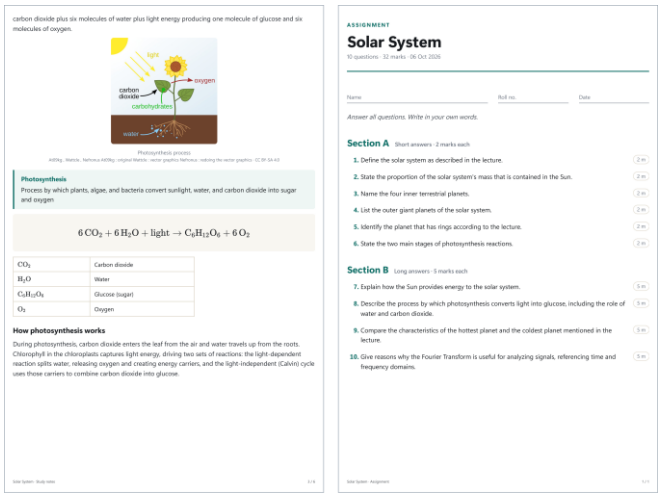
object the LLM suggests a query; an openly licensed Wikimedia image (source, author and licence kept) is placed only if CLIP's zero-shot similarity [16] ranks it above generic distractor prompts ("a logo", "a map") and the query matches the slide text.

### G. Live Display and Teacher Control

A FastAPI server pushes versioned slide patches over WebSocket to /display (projector), /control (teacher) and /view (students, shared link). The control view (Fig. 2) adds editing, image search and upload, pause, blank screen, mistake cards, the running transcript and the lecture structure tree.

### H. Lecture Materials

After the lecture the event log is reused: on request it becomes study notes, an assignment (Fig. 3), a PPTX deck, or a live quiz or test answered through the shared link, all from the taught content and edited by the teacher. These features are not evaluated here.




(a) Study notes (b) Assignment

Fig. 3. Materials generated from the event log of a recorded D1 lecture (not evaluated): (a) a study-notes page; (b) the assignment.

## IV. Experimental Evaluation

### A. Setup and Data

Every run used one laptop (NVIDIA RTX 3050, 4 GB video memory; 16 GB system memory). D1 is six lectures spoken by the authors (114.1 min). The held-out set D2 (Tables II and III) has 30 lectures spoken live, unscripted, by six speakers into the laptop microphone (Recorded, physics to economics) and six 10-min public lectures played in as audio in real time (Public): MIT OpenCourseWare 7.016, 5.111, 18.01 and 8.01 (CC BY-NC-SA) and NPTEL Computer Networks and History of Economic Theory (CC BY-SA). The speakers consented to the recording. Against MIT's human captions of the 7.016 lecture (1,257 words), the live transcript has a word error rate of 7.1 %. Teachers did not edit slide text during a test; in D2 they deleted two items and accepted corrections from concern cards in two lectures. Three populations are evaluated: all 36 lectures for latency and operation, an annotated subset of 18 (12 Recorded, 6 Public; 191 min) for accuracy and the ablation, and 30 sampled slides for teacher ratings.

TABLE II
Evaluation Data

| Set | Lectures | Minutes | Units | Fallback units | Slides |
|---|---|---|---|---|---|
| D1 (development) | 6 | 114.1 | 389 | 45 | 163 |
| D2 Recorded | 30 | 349.2 | 1,275 | 59 | 453 |
| D2 Public | 6 | 60.0 | 344 | 13 | 52 |
| **D2 all (held out)** | **36** | **409.2** | **1,619** | **72** | **505** |

Units: groups of finished sentences sent for interpretation; fallback units were interpreted without the LLM (quota spent or provider failure).

TABLE III
Live System Performance (Session Logs)

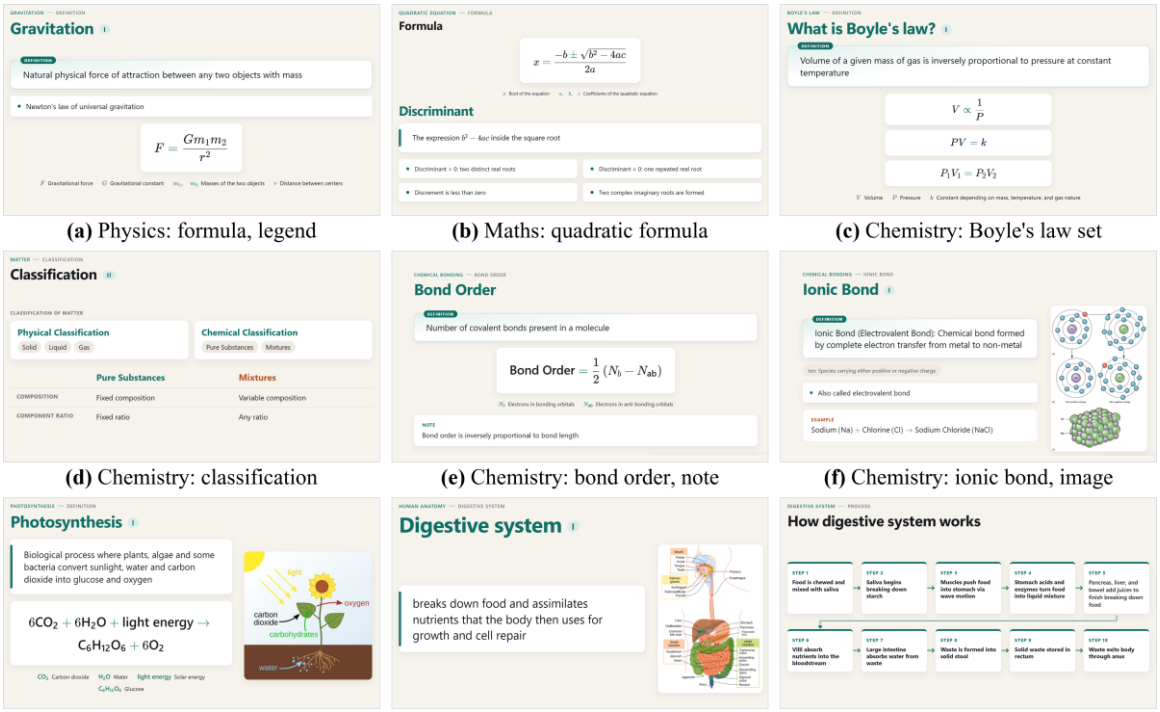
| Measure (s) | D1 | D2 Rec. | D2 Pub. | D2 all |
|---|---|---|---|---|
| Speech-to-text per utterance | 0.66 | 0.71 | 0.60 | **0.68** |
| LLM call | 1.35 | 1.92 | 2.23 | **1.96** |
| Speech to slide, median | 12.5 | 10.1 | 17.1 | **11.2** |
| Speech to slide, 95th percentile | 36.8 | 23.2 | 93.4 | **53.0** |

Medians unless stated; speech to slide: from the end of a spoken line to its display. Public includes web4 and web6 (provider outages).

*Annotation.* Two authors (A and B) independently annotated the 18 lectures, chosen to mix subjects and speakers with 6–17 min each. Before seeing the slides, each marked topic boundaries (where a teacher would start a new board) and key facts in the transcript, then judged whether each key fact was on a slide and whether each slide statement was said and correct; scores are the mean of A and B, without adjudication. A system boundary is the utterance that opened a new content slide. Boundaries are scored by precision, recall and F1 within ±1 utterance, averaged per lecture, and by $P_k$ [24], the probability that two utterances half a segment apart are wrongly split or joined; agreement is Cohen's κ [29] and, for skewed labels, Gwet's AC1 [30]. Intervals are 95 % bootstrap intervals over lectures. *Ratings.* Five teachers (T1–T5), not involved in development, rated 30 random D2 slides, each shown with what was said, from 1 to 5 on four criteria; their consistency is the ICC(2,1) of Shrout and Fleiss [31].

### B. Topic Boundaries and Content Accuracy

Slides change where the topic changes (Table IV): boundary precision is 0.78 (95 % interval 0.68–0.88), so a projected slide is rarely replaced within a topic, and $P_k$ is 0.32. Recall is 0.42 (0.30–0.55): a new concept is not a new slide, so the planner merges into the live slide sub-points that annotators marked as boundaries, most often in the Public lectures (recall 0.19); Section IV-C shows this is a trade-off. Of 669 key facts, 86.6 % (84.6–88.5 %) reached a slide (partly shown counts ½), and only 0.9 % (0.2–1.8 %) of 2,504 statement judgments were not supported by the speech. The coverage guard (3) is needed for this: 60 of the covered facts reached a slide only through a sentence it restored, so without it coverage would be 77.7 %; the guard adds 9.0 points (6.4–11.7). Of the 3.7 % marked wrong, 41 of 79 were also marked as said (a slip or mis-hearing): slides follow the speech, so transcript support is not factual correctness. Agreement is good for boundaries (κ = 0.72) and high for "supported" (88.8 %, AC1 0.88); κ is only 0.07 there because nearly all statements were supported, which lowers κ but not AC1 [30].




**(a)** Physics: formula, legend **(b)** Maths: quadratic formula **(c)** Chemistry: Boyle's law set

**(d)** Chemistry: classification **(e)** Chemistry: bond order, note **(f)** Chemistry: ionic bond, image

**(g)** Biology: reaction, image **(h)** Biology: labelled image **(i)** Biology: ten-step process

Fig. 4. Slides from development lectures by the authors; (a), (c), (i) re-render recorded LLM answers, the others are app screenshots. Text is as spoken (the slip "Discrement" in (b)).

Test29 ran entirely on the fallback (quota spent). It still covered 83.1 % of key facts with 0.7 % unsupported statements, but it rarely opened a slide (recall 0.04) and 11 of its 13 slides were plain points, so the fallback keeps the content and the LLM adds the structure. The 17 LLM lectures reach precision 0.77 and recall 0.45. Teachers rated the slides 3.87–4.17 of 5 (ICC(2,1) 0.45–0.56, moderate). In Fig. 4, the format is determined by the type of information contained rather than the topic.

TABLE IV
ACCURACY AND TEACHER RATINGS (18 ANNOTATED D2 LECTURES)

| Measure | Recorded | Public | All |
|---|---|---|---|
| *Topic boundaries* | | | |
| Precision ↑ | 0.81 | 0.73 | **0.78** |
| Recall ↑ | 0.54 | 0.19 | **0.42** |
| F1 ↑ | 0.59 | 0.29 | **0.49** |
| $P_k$ ↓ | 0.29 | 0.37 | **0.32** |
| *Slide content* | | | |
| Key facts on a slide ↑ | 88.0 % | 84.0 % | **86.6 %** |
| without coverage guard | 81.9 % | 69.1 % | 77.7 % |
| Statements not said ↓ | 0.9 % | 0.9 % | **0.9 %** |
| Statements marked wrong ↓ | 3.6 % | 4.0 % | **3.7 %** |
| *Teacher ratings (1–5)* ↑ | | | |
| Correctness | 4.20 | 4.06 | **4.15** |
| Relevance | 3.94 | 3.80 | **3.89** |
| Readability | 4.12 | 4.26 | **4.17** |
| Usefulness | 3.93 | 3.76 | **3.87** |
| *Annotator agreement* | | | |
| Boundaries, κ (raw) | 0.81 (94 %) | 0.56 (90 %) | 0.72 (92 %) |
| Statement said, AC1 (raw) | 0.89 (90 %) | 0.84 (85 %) | 0.88 (89 %) |

↑ / ↓: higher / lower is better. Mean of annotators A and B; boundaries within ±1 utterance; ratings: T1–T5 mean over 30 slides.

## C. Ablation of the Planner Rules

In Table V, the recorded signals and LLM answers of the 18 lectures are replayed through the planner with one rule changed at a time, without LLM requests; the unchanged replay gives the same decks as the live run. The continuity rule matters most: when every concept gets its own slide, the deck changes 3.2 times per minute, 200 slides are replaced within 10 s of being shown, and precision drops to 0.37 (Pk = 0.47). The dwell time only delays a change: without it the changes per minute stay at 1.15, but slides replaced within 10 s go up from 16 to 25. With a dwell time of up to 20 s, precision and recall do not change; at 45 s they drop to 0.67 and 0.32. Topic confirmation made no measurable difference on D2 (precision 0.79 without it) and is kept only as a safeguard. When the LLM labels decide but slides are still only extended, the boundaries hardly change, so the stability comes from the extend-only operations, not from the labels.

TABLE V
PLANNER ABLATION BY REPLAY (18 ANNOTATED D2 LECTURES)

| Planner variant | Precision ↑ | Recall ↑ | $P_k$ ↓ | Changes per min | Replaced < 10 s ↓ |
|---|---|---|---|---|---|
| **Full planner** (=live) | **0.78** | 0.42 | **0.32** | **1.15** | 16 |
| No dwell time | 0.78 | 0.42 | 0.32 | 1.15 | 25 |
| No topic confirmation | 0.79 | 0.45 | 0.31 | 1.12 | 10 |
| LLM labels decide | 0.78 | 0.46 | 0.30 | 1.17 | 21 |
| Each concept a slide | 0.37 | 0.86 | 0.47 | 3.18 | 200 |

Boundary scores as shown in Table IV; replaced < 10 s: slides replaced within 10 seconds of their presentation.

### D. Comparison With an LLM-Controlled Baseline

TABLE VI
PROPOSED SYSTEM VS. LLM-CONTROLLED BASELINE (SAME D2 UNITS)

| Measure | Proposed | LLM baseline |
|---|---|---|
| Seen statements rewritten/removed ↓ | **2.0 %** | 38.2 % |
| Seen statements lost from final deck ↓ | **6.2 %** | 65.8 % |
| Lectures with fewer rewrites ↑ | **18 of 18** | 0 of 18 |
| Boundary precision ↑ | **0.78** | 0.72 |
| *Pk* ↓ | **0.32** | 0.36 |
| Content slides per minute | 1.25 | 0.79 |

↑ / ↓: higher / lower is better. Rewritten: statements of a replaced slide version that the next version changed or removed; lost: shown statements absent from the final deck.

An LLM-controlled baseline without extend-only operations receives, for the same 850 units of the 18 annotated lectures, the slide it showed last and the last 60 s of transcript, and returns the slide to show, without planner, guards or composer (same two models, 0.5 M tokens). It placed slide changes about as well (Table VI; precision 0.72 vs. 0.78, $p = 0.34$) but rewrote or removed 1,439 of the 3,763 statements students had already seen (38.2 %), against 91 of 4,515 (2.0 %), and 65.8 % of what it showed was gone from its final deck, against 6.2 %. The difference holds in all 18 lectures: the mean per-lecture gap in rewrites is 39 points (95 % interval 32–47; exact paired sign-flip permutation test, $p < 0.001$). Of the proposed system's 91 changes, 24 extended a statement, 6 followed a correction the teacher accepted and 61 were composer merges. The baseline also lacks the guards and composer; Table V isolates the planner rules.

### E. End-to-End Latency

Over the 36 D2 lectures (Table III), planning and composing take 2 ms; the median from the end of a spoken line to its display is 11.2 s, 7.6 s of it the gate's deliberate wait for a complete idea: the system is live, not hard real-time. In web4 and web6 the free provider failed (timeout, HTTP 429), and these two lectures are included. With them the 95th percentile is 53.0 s; without them the median is 10.2 s and the 95th percentile 23.4 s.

## V. DISCUSSION AND CONCLUSION

With the LLM as a bounded interpreter, the copilot builds a continuity-aware deck from live speech, and extend-only operations keep what students read stable (2.0 % vs. 38.2 % rewritten). Limitations: the annotators were the authors (though the annotations were done before viewing the slides); recall is moderate by design; an "append only" baseline prompt was not tested; WER is calculated from the single lecture that has human captions; the free LLM provider caused outages; no learning effect is claimed since no classroom tests were done. Anonymized annotations, ratings and evaluation scripts are available on request.

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